

DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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Football Match Manager - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Football Match Manager is a RESTful API designed to manage football matches, teams, and leagues, featuring a user-following system for specific matches. The system implements a JWT-based authentication pattern and a versioned routing structure (`/api/v1`) to ensure backward compatibility and secure access to personalized user data.

1.2 Maturity Assessment

The project is in a REFINEMENT state. While the API contract is technically complete with a high completeness score, it lacks critical business context. The absence of high-priority 'Goals & Objectives' and 'Scope' definitions creates a strategic vacuum, meaning the technical implementation is ready but the product direction is not yet anchored.

1.3 Technical Stack

- **API Architecture:** RESTful
- **Data Format:** JSON
- **Authentication:** JWT (JSON Web Token)
- **Identifier Standard:** UUID

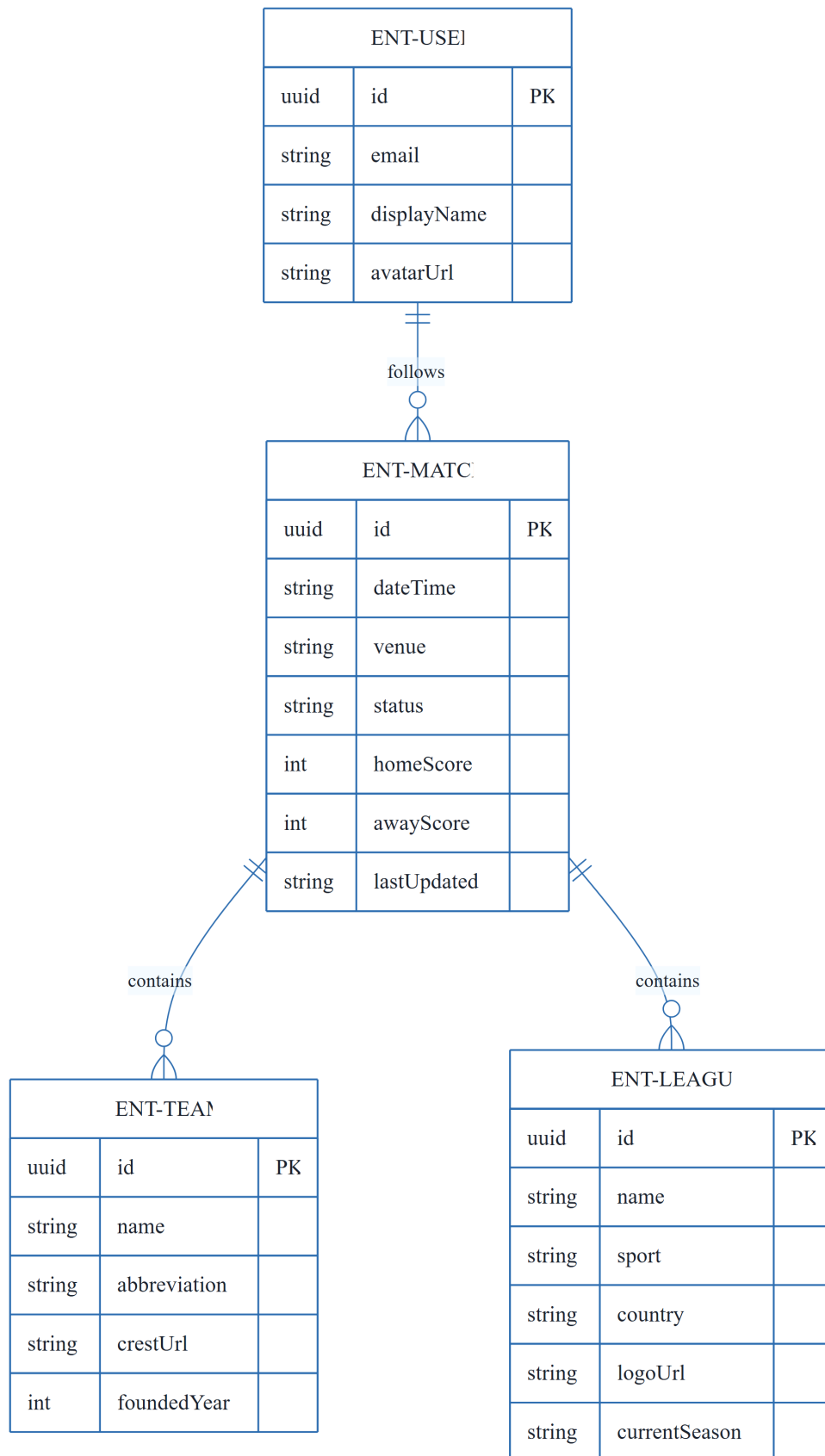
1.4 Architectural Constraints

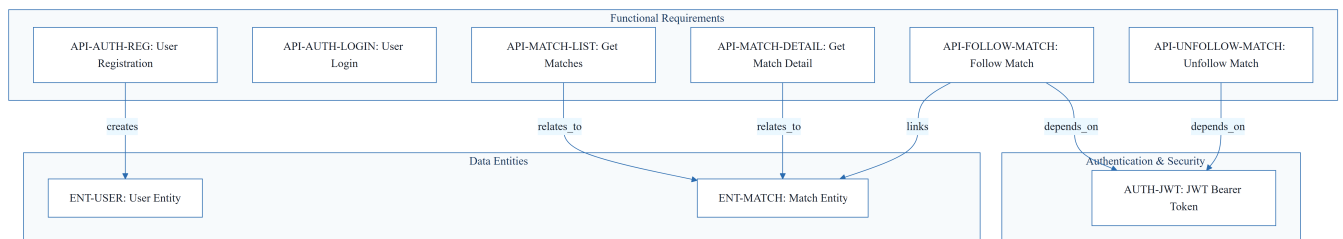
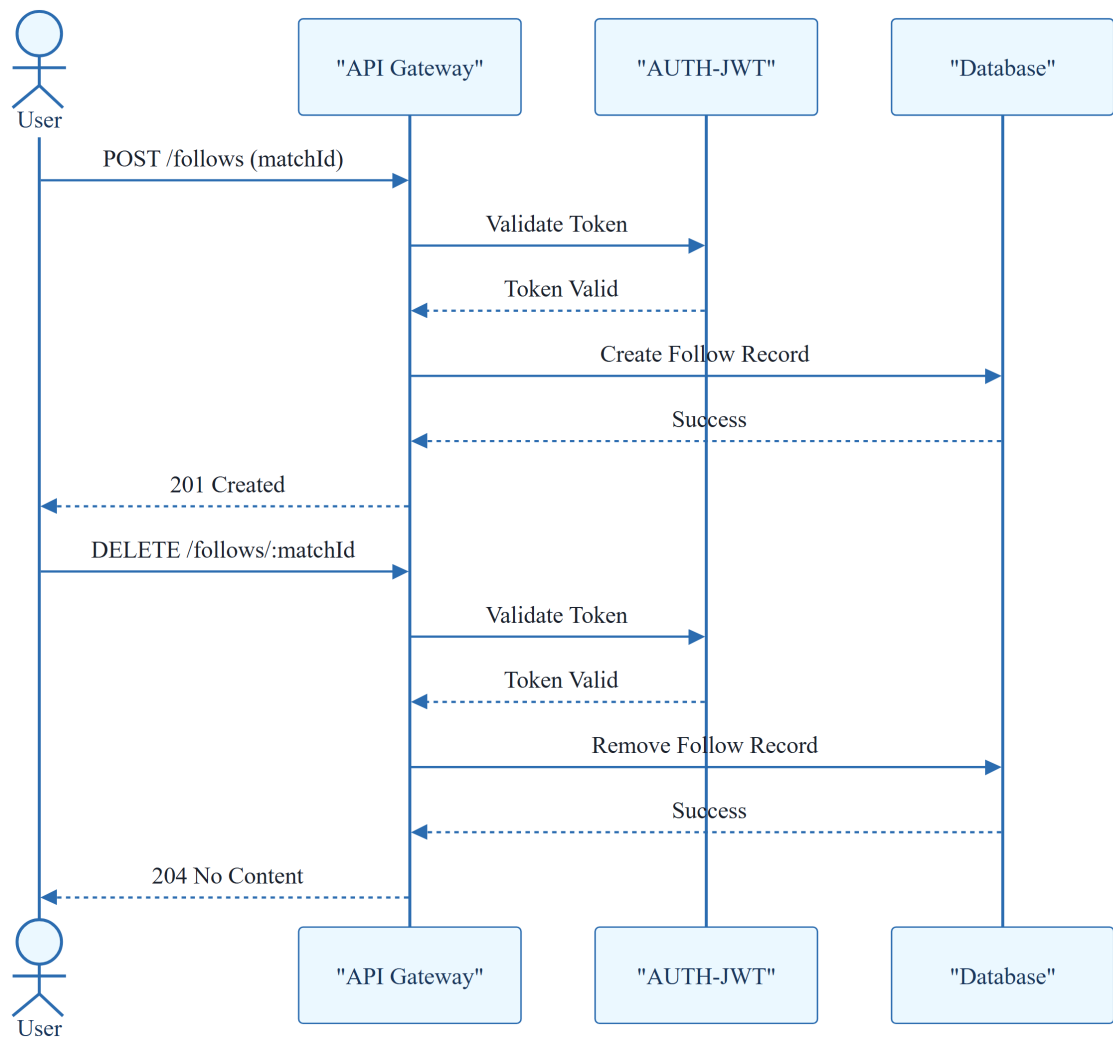
- **API Versioning:** Mandatory URL path prefix `/api/v1/`.
- **Rate Limiting (Auth):** Maximum 5 requests per minute per IP.
- **Rate Limiting (General):** Maximum 60 requests per minute per user.
- **Authentication:** Bearer Token (JWT) required for all non-auth endpoints.
- **Pagination Defaults:** Matches list default page 1, limit 20.
- **Teams List Default:** Limit 50 results.

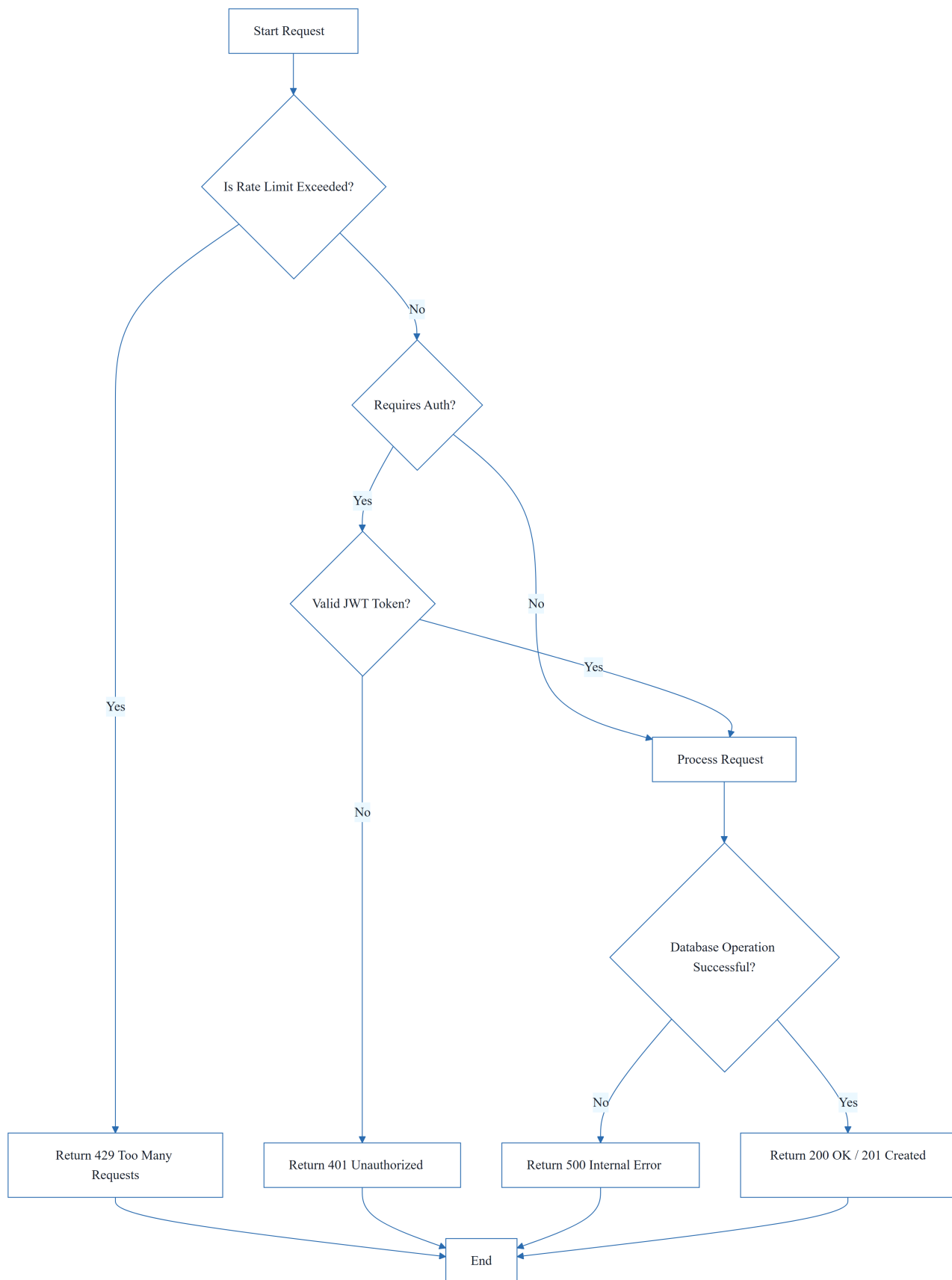
1.5 Critical Dependencies

- JWT for secure session management and access control.
- Referential integrity between ENT-MATCH, ENT-TEAM, and ENT-LEAGUE.
- User-Match relationship for the following functionality.
- ISO string format for date-based match filtering (`dateFrom`, `dateTo`).

2. Architecture Workflows

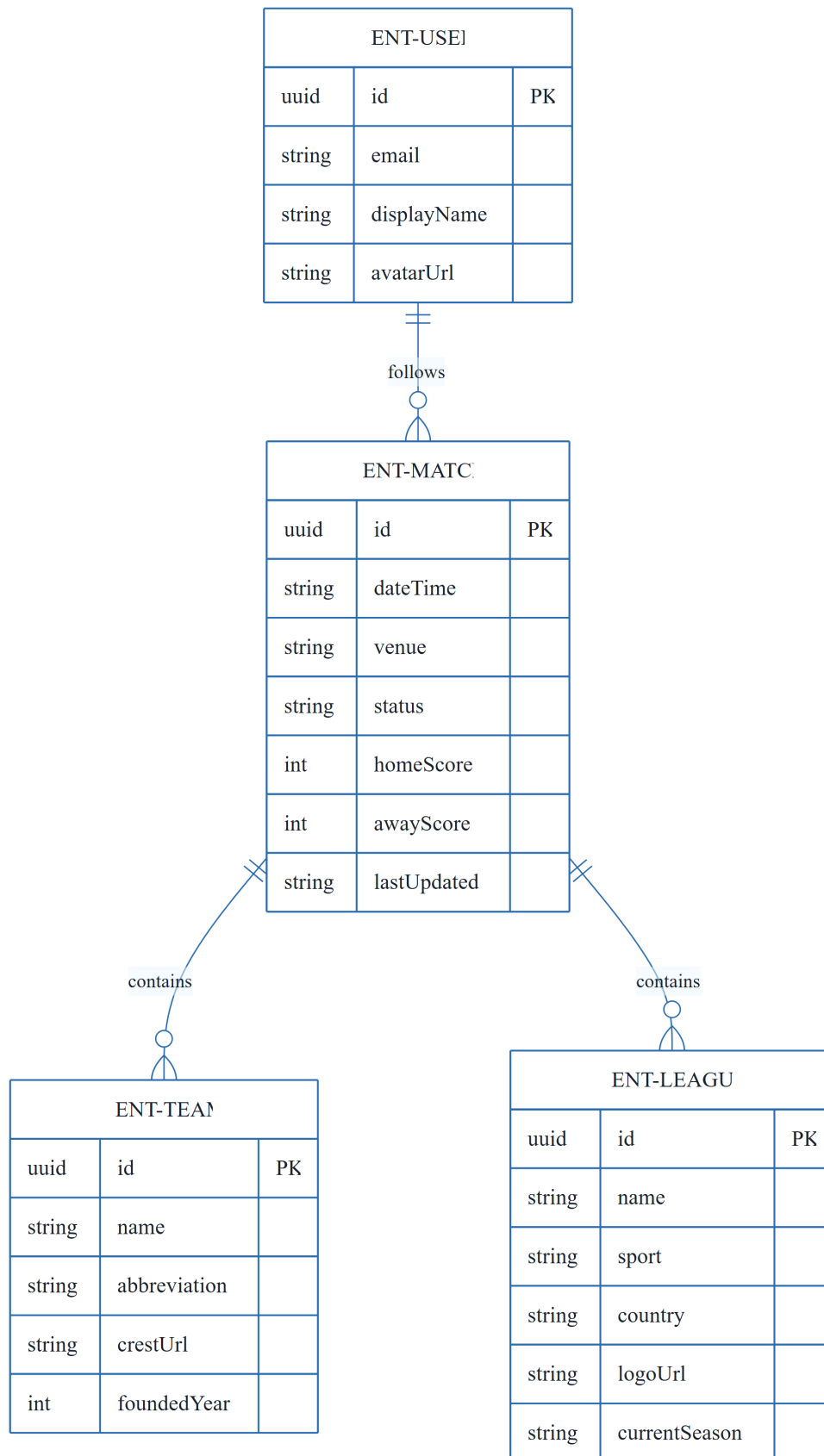




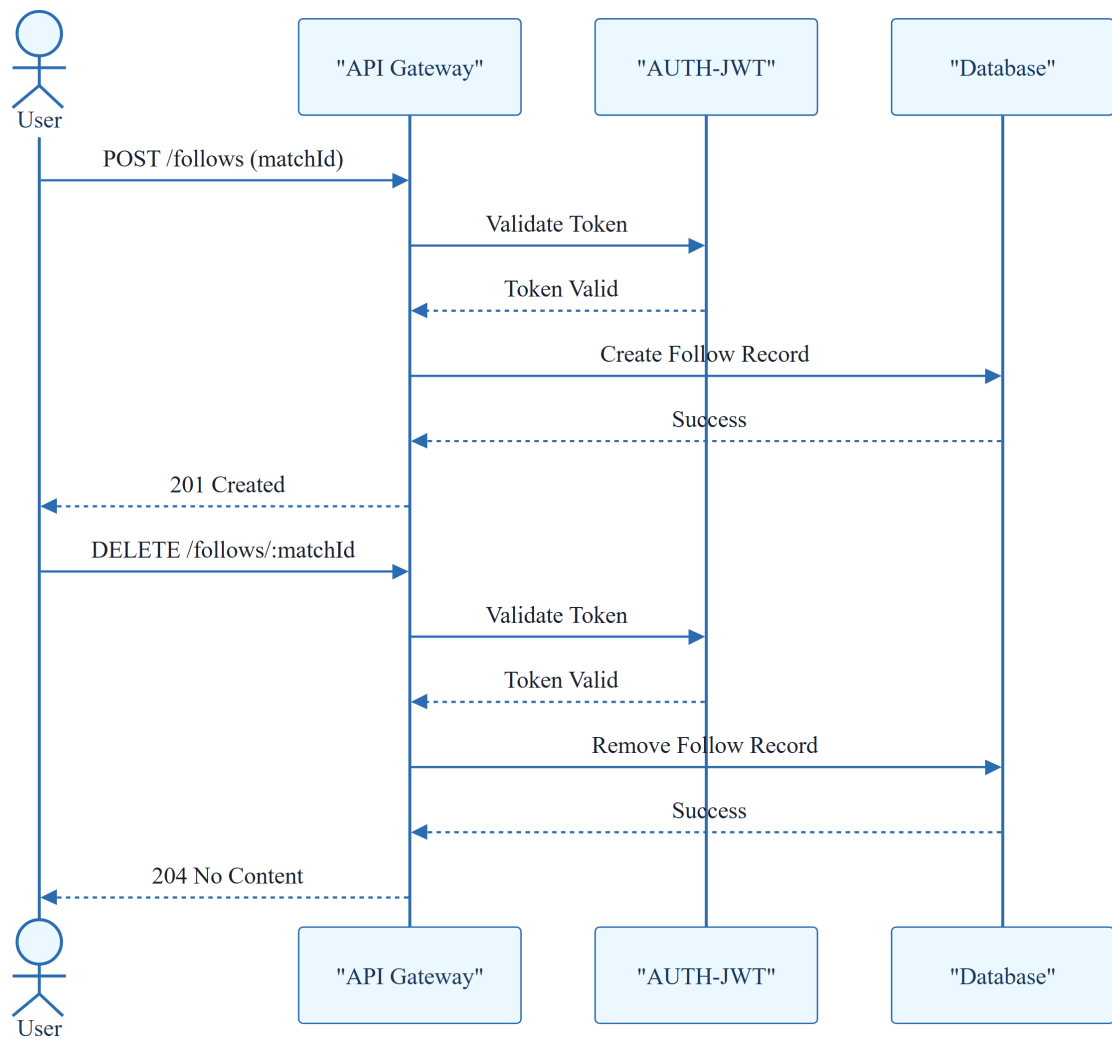


& Visual Diagrams

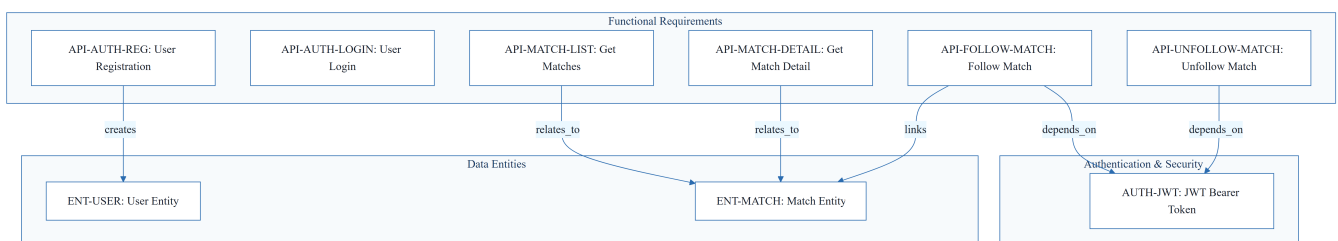
2.1 Football Match Manager Data Model



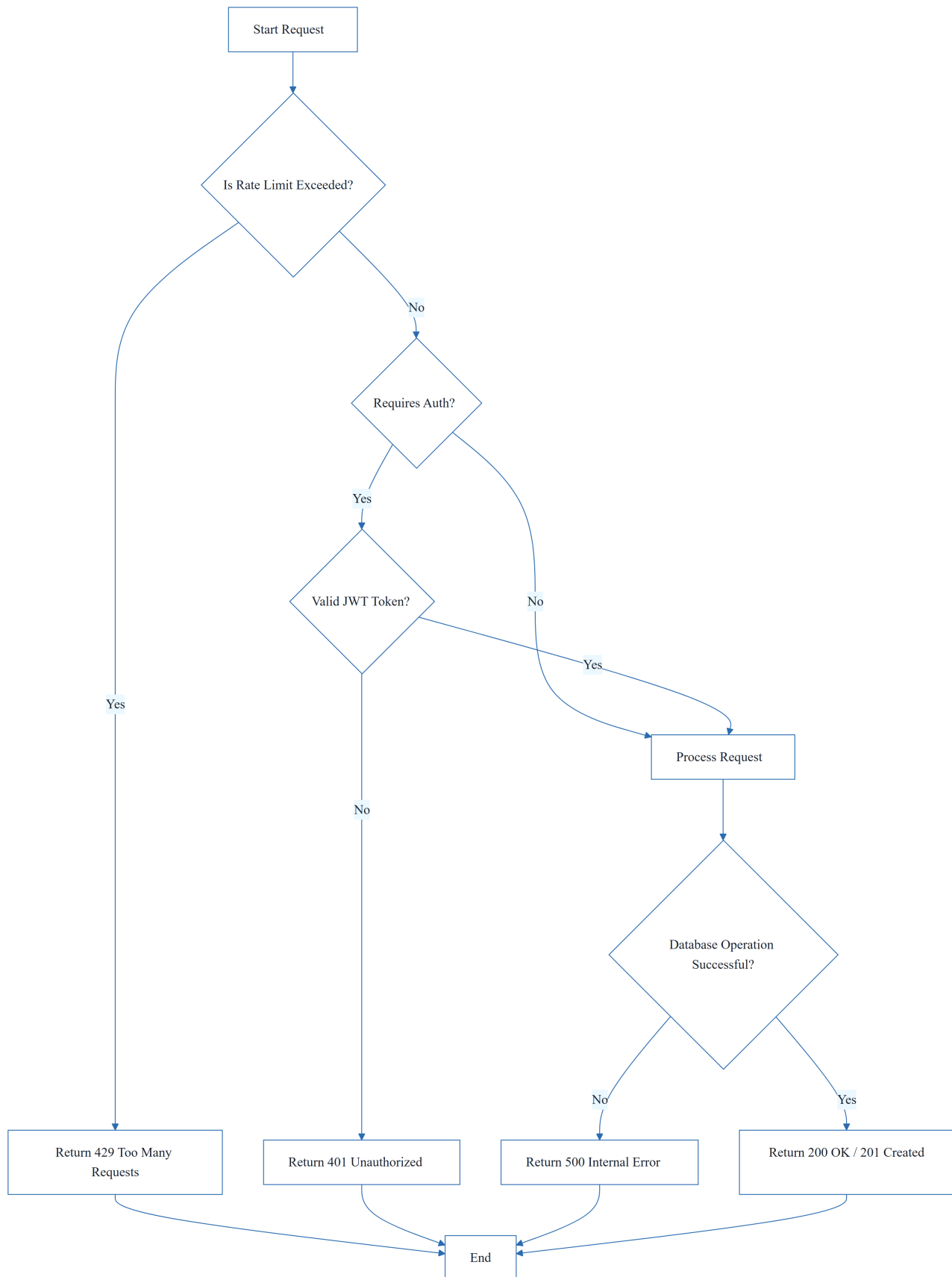
2.2 Match Following Sequence



2.3 Requirements Traceability Map



2.4 API Request Processing Workflow



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Type	Requirement / Entity Description	Source Section
AUTH - JWT	NFR	All endpoints except auth endpoints require a valid JWT token in the Authorization header.	Authentication
API - AUTH - REG	FR	Allow new users to register via POST /auth/register.	Authentication
API - AUTH - LOGIN	FR	Allow users to login and receive a JWT token via POST /auth/login.	Authentication
API - MATCH - LIST	FR	Retrieve a list of matches with filtering by leagueId, date range, status, and team search, including pagination.	Matches
API - MATCH - DETAIL	FR	Retrieve a specific match by its unique ID.	Matches
API - FOLLOW - MATCH	FR	Allow an authenticated user to follow a match via POST /follows.	User Follows
API - UNFOLLOW - MATCH	FR	Allow an authenticated user to unfollow a match via DELETE /follows/:matchId.	User Follows
ENT - MATCH	Entity	Match entity containing homeTeam, awayTeam, dateTime, venue, league, status, and scores.	Match
ENT - TEAM	Entity	Team entity containing name, abbreviation, crestUrl, and foundedYear.	Team
ENT - LEAGUE	Entity	League entity containing name, sport, country, logoUrl, and currentSeason.	League
ENT - USER	Entity	User entity containing email, displayName, and avatarUrl.	User (authenticated user only)
NFR - RATE - LIMIT	NFR	Auth endpoints limited to 5 req/min per IP; other endpoints limited to 60 req/min per user.	Rate Limiting
CON - VERSIONING	Constraint	API versioning must be handled via URL path (/api/v1/).	Versioning

3.2 Security Rules

- **Authentication Mechanism:** Bearer Token (JWT).
- **Access Control:** All endpoints except /auth/* require a valid JWT.
- **Rate Limiting:**
 - Authentication endpoints: 5 requests/minute per IP.
 - General endpoints: 60 requests/minute per authenticated user.
- **Error Handling:** Standardized error response body: { "error": "string", "message": "string", "details": "object (optional)" }.

3.3 Data Models

ENT-MATCH

```
{
  "id": "uuid",
  "homeTeam": { "id": "uuid", "name": "string" },
  "awayTeam": { "id": "uuid", "name": "string" },
  "dateTime": "ISO string",
  "venue": "string",
  "league": { "id": "uuid", "name": "string" },
  "status": "upcoming|live|finished|postponed|cancelled",
  "homeScore": "integer (nullable)",
  "awayScore": "integer (nullable)",
  "lastUpdated": "ISO string"
}
```

ENT-TEAM

```
{
  "id": "uuid",
  "name": "string",
  "abbreviation": "string",
  "crestUrl": "url",
  "foundedYear": "integer"
}
```

ENT-LEAGUE

```
{
  "id": "uuid",
  "name": "string",
  "sport": "string",
  "country": "string",
  "logoUrl": "url",
  "currentSeason": "string"
}
```

ENT-USER

```
{
  "id": "uuid",
  "email": "string",
  "displayName": "string",
  "avatarUrl": "url"
}
```

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Goals & Objectives	HIGH	The document is purely technical. Business goals, target audience, and the 'Why' behind the API are missing.
Scope & Out-of-Scope	MEDIUM	Define what the API will NOT handle (e.g., real-time socket updates, payment processing).
Open Questions & Uncertainties	LOW	Identify any pending decisions regarding the API contract or data models.

4.2 Remediation & Workflow

To move from the REFINEMENT state to a PRODUCTION-READY state, the project lead must define the business objectives and scope. Once the "Why" is anchored, the technical specifications can be validated against these goals to ensure no functional gaps exist in the API contract.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
API	TECHNICAL_STACK	API Contract: Football Match Manager	The RESTful interface providing endpoints for match, team, and league management.
CORS Standard	TECHNICAL_STACK	Error Responses	The browser security mechanism governing cross-origin resource sharing for the backend endpoints.

Term	Category	Context Anchor	Project Definition
ID	TECHNICAL_STACK	ENT-MATCH	The primary unique identifier for system entities.
IP	TECHNICAL_STACK	NFR-RATE-LIMIT	The network address used to track and limit authentication requests to 5 per minute.
JSON	TECHNICAL_STACK	Data Models	The lightweight data-interchange format used for all request and response bodies.
JWT	TECHNICAL_STACK	AUTH-JWT	The signed credential passed in the Authorization header to secure non-auth endpoints.
UUID	TECHNICAL_STACK	ENT-USER	The 128-bit universally unique format used for all entity primary keys.
dateFrom	BUSINESS_DOMAIN	API-MATCH-LIST	The lower temporal boundary for filtering matches using an ISO string.
dateTo	BUSINESS_DOMAIN	API-MATCH-LIST	The upper temporal boundary for filtering matches using an ISO string.
leagueId	BUSINESS_DOMAIN	API-MATCH-LIST	The specific identifier used to filter match results by a particular competition.